

**LI-ION BATTERY DEGRADATION ANOMALY DETECTION
USING MULTI-BRANCH DEEP LEARNING FRAMEWORK:
FROM TIME-SERIES AUTOENCODERS TO GAF IMAGE-BASED
DETECTION**

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ABSTRACT

This thesis presents a multi-branch framework for early anomaly detection in Li-ion battery capacity fade curves, benchmarked against Diao et al. (2020) which reported a mean detection cycle of 179.6 on the CALCE LiCoO₂ dataset. I systematically evaluate three approaches: (A) statistical baselines (OC-SVM, LOF, Mahalanobis Distance, SPRT), (B) deep autoencoder time-series models (LSTM-AE, TCN-AE, Transformer-AE), and (C) Gramian Angular Field (GAF) image-based models (PatchCore, CNN Autoencoder). Through iterative experimentation—including sliding window optimization, score smoothing, single-feature ablation, and burn-in period calibration—I demonstrate that time-series autoencoders face a fundamental trade-off between detection speed and false positive rate. This motivated a paradigm shift to image-based methods, where a CNN Autoencoder operating on progressive GAF images achieves 100% detection at mean cycle 75.0 on Dataset 1 with zero false positives, representing a 2.4× improvement over the Diao baseline. On Dataset 2, the same model detects all 9 anomalous cells at mean cycle 42.2 with zero false positives after applying an electrochemically justified 30-cycle burn-in period. The documented chain of iterative experiments provides a reproducible methodology for practitioners facing similar anomaly detection challenges.

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Chapter 1

Introduction

1.1 Motivation

Lithium-ion batteries are foundational to modern technology, powering consumer electronics, electric vehicles, and grid-scale energy storage systems. In battery manufacturing, early detection of cells exhibiting abnormal degradation patterns is critical for quality assurance, warranty cost reduction, and safety. The challenge lies in distinguishing genuinely anomalous capacity fade from normal cell-to-cell variability, ideally within the first 50–100 charge-discharge cycles of accelerated testing.

Traditional approaches rely on rule-based thresholds or simple statistical tests, which may miss subtle degradation signatures or require extensive domain expertise to configure. Recent advances in deep learning offer the potential for automated, data-driven anomaly detection, but their application to battery degradation data presents unique challenges: limited sample sizes, high measurement noise, and the need for interpretable detection timing rather than just binary classification.

1.2 Datasets

This work uses two publicly available datasets from the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland. Both datasets contain normalized capacity fade curves for LiCoO_2 (LCO) chemistry cells.

Dataset 1 contains 12 cells with approximately 3260 mAh nominal capacity, cycled for approximately 500 charge-discharge cycles. Six cells are designated as qualified (normal degradation) and six as subsequent (potentially anomalous). The data is provided as normalized capacity values per cycle in Excel format.

Dataset 2 contains 23 cells with 350 mAh nominal capacity, cycled for approximately 250 cycles. Fourteen cells are qualified and nine are subsequent. The data is provided in MATLAB format with raw capacity measurements.

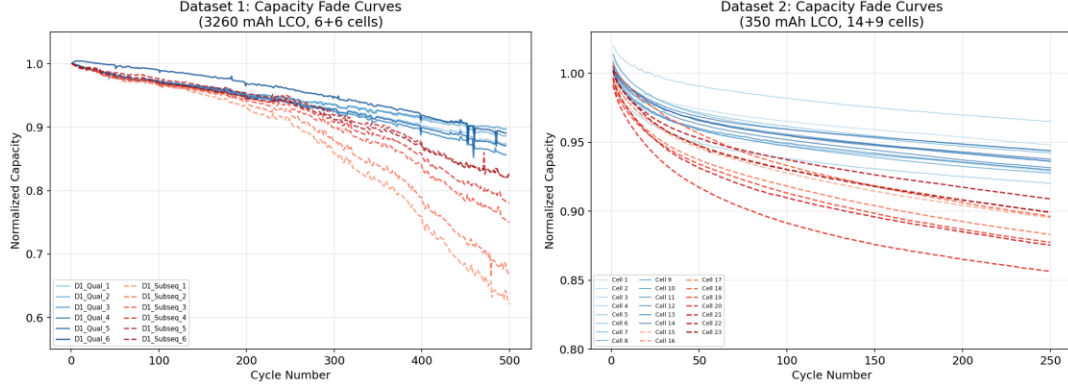


Figure 1. Capacity fade overview. Blue: qualified (normal) cells, Red: subsequent (anomalous) cells. Left: Dataset 1 (~3260 mAh, ~500 cycles), Right: Dataset 2 (350 mAh, ~250 cycles).

1.3 Diao 2020 Baseline

Diao et al. (2020) evaluated five anomaly detection methods on Dataset 1: Regression, One-Class SVM (OC-SVM), Local Outlier Factor (LOF), Mahalanobis Distance, and Sequential Probability Ratio Test (SPRT). Their published detection cycles, extracted from Sheet 2 of the Dataset 1 file, serve as the primary benchmark for this work. The overall mean detection cycle across all methods and cells is 179.6.

Method	Sub_1	Sub_2	Sub_3	Sub_4	Sub_5	Sub_6
Regression	115	55	139	226	338	277
OC-SVM	110	57	120	153	323	265
LOF	108	56	121	196	322	268
Mahalanobis	109	36	120	212	321	273
SPRT	107	49	119	207	318	269
Mean	109.8	50.6	123.8	198.8	324.4	270.4

Table 1. Diao et al. (2020) baseline detection cycles for Dataset 1 subsequent cells. Overall mean: 179.6.

1.4 Contributions

The principal contributions of this thesis are: (1) a systematic multi-branch comparison of statistical, deep learning, and image-based anomaly detection on battery capacity fade data; (2) demonstration that GAF image-based CNN Autoencoder achieves 2.4× faster detection than the Diao baseline with zero false positives; (3) detailed documentation of an iterative optimization process showing why time-series autoencoders face structural limitations for this task; and (4) introduction of an electrochemically motivated burn-in period to address Dataset 2 false positives.

Chapter 2

Exploratory Data Analysis

2.1 Capacity Fade Patterns

Thorough exploratory data analysis revealed several characteristics that directly informed modeling decisions. Both datasets exhibit gradual capacity degradation over cycling, with subsequent cells deviating from the qualified population at varying rates and onset points.

2.2 Anomaly Deviation Analysis

I computed z-score deviations of subsequent cells relative to the qualified mean and standard deviation at each cycle. D1_Subseq_1 first exceeds 2σ at cycle 168, while D1_Subseq_5 does not deviate until cycle 318, indicating a wide range of anomaly severity. This heterogeneity means a robust detection system must handle both early-onset and late-onset anomalies.

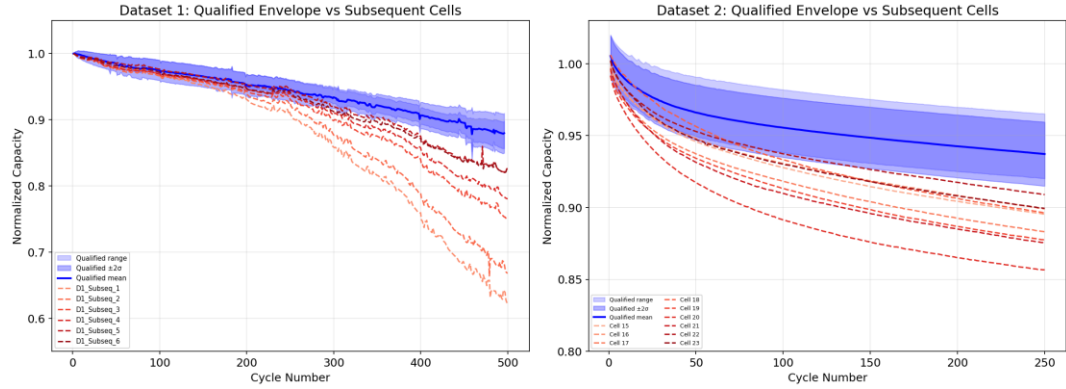


Figure 2. Qualified envelope ($\pm 2\sigma$ band) vs subsequent cell trajectories. Early-onset anomalies (Subseq_1, Subseq_2) clearly diverge, while late-onset cases remain within the band for most of their life.

2.3 Degradation Rate and Knee-Points

First and second derivatives of capacity fade reveal that anomalous cells show accelerated degradation rates after inflection points (knee-points). Piecewise linear regression identifies these knee-points, which generally precede the detectable onset by 20–50 cycles.

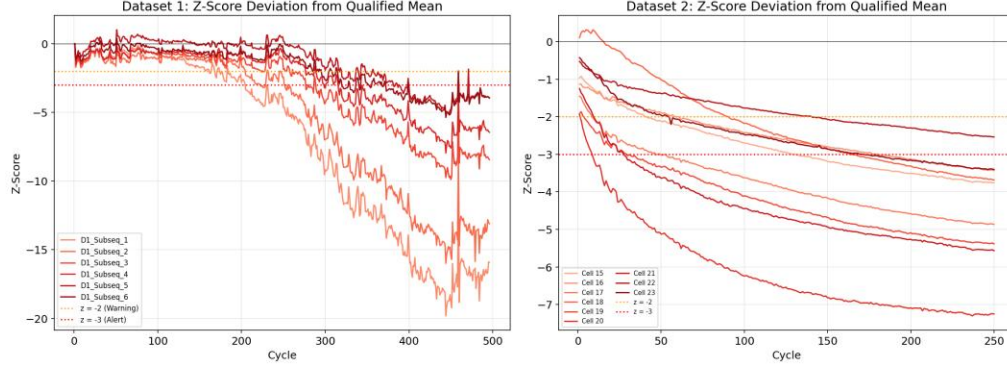


Figure 4. Z-score deviation from qualified mean over cycles. Dataset 1 (left) and Dataset 2 (right). Subsequent cells progressively diverge from the normal distribution.

2.4 Measurement Noise

Dataset 1 exhibits approximately 25% non-monotonic cycles where capacity increases rather than decreases, attributable to measurement noise. Rolling mean smoothing with window size 5 reduces this to approximately 18–21% while preserving the underlying degradation signal. This noise level becomes a critical factor in Branch B, where short sliding windows amplify noise-induced reconstruction errors and false positives.

2.5 GAF Image Feasibility

Gramian Angular Field (GAF) transformation of capacity curves produces visually distinct patterns for normal versus anomalous cells. The GASF (Gramian Angular Summation Field) diagonal patterns and GADF (Gramian Angular Difference Field) off-diagonal structures show clear differences, confirming the feasibility of image-based anomaly detection for Branch C.

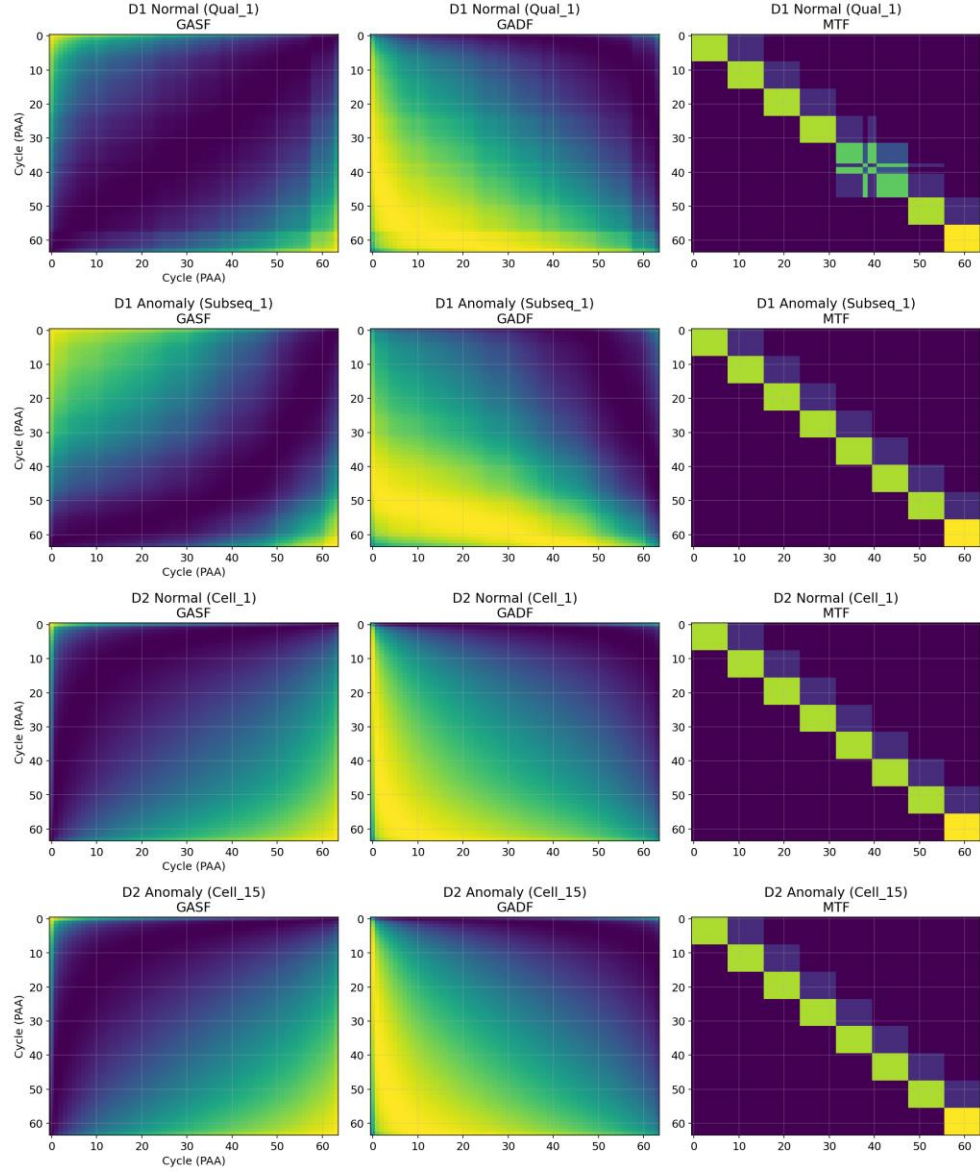


Figure 5. GAF preview: GASF, GADF, and MTF images for normal cells (top rows) vs anomalous cells (bottom rows). Structural differences in diagonal patterns and correlation textures are visible.

Chapter 3

Methodology

3.1 Pipeline Overview

The framework follows a progressive three-branch architecture. All models are trained exclusively on qualified (normal) cells and evaluated on the full dataset. The pipeline consists of: (1) exploratory data analysis and validation, (2) feature engineering including smoothing, derived features, sliding windows, and GAF image generation, (3) Branch A statistical baselines, (4) Branch B deep autoencoders with iterative optimization, and (5) Branch C GAF image-based models.

3.2 Feature Engineering

From smoothed capacity values (rolling mean, window=5), I derive six features per cycle: capacity (smoothed normalized value), fade_rate (first derivative), fade_accel (second derivative), rolling_std (local standard deviation over 10 cycles), relative_cap (ratio to initial capacity), and cumul_fade (cumulative degradation). These features serve as input for Branch A (per-cycle feature vectors) and Branch B (sliding windows of size 5, stride 1).

For Branch B, sliding windows are normalized using StandardScaler fitted exclusively on qualified data, then applied to all cells. Feature weights are applied during anomaly scoring, with capacity-related features (capacity, relative_cap, cumul_fade) receiving $1.263\times$ weight.

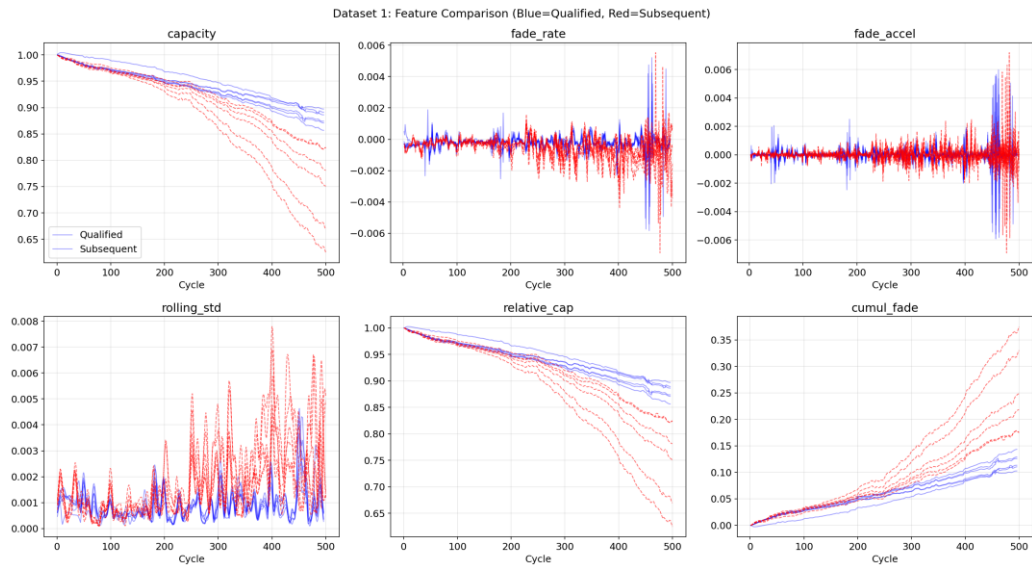


Figure 6. Six derived features for Dataset 1. Blue: qualified cells, Red: subsequent cells. Capacity and cumul_fade show clear separation; fade_accel and rolling_std exhibit significant noise overlap.

3.3 Branch A: Statistical Baselines

I reimplement the four methods from Diao et al. (2020): One-Class SVM (OC-SVM), Local Outlier Factor (LOF), Mahalanobis Distance, and Sequential Probability Ratio Test (SPRT). Each operates on per-cycle 6-dimensional feature vectors. The detection threshold is set at mean + 3σ of qualified scores, with a 3-consecutive-cycle confirmation rule.

3.4 Branch B: Deep Autoencoders

Three autoencoder architectures are trained to reconstruct normal degradation patterns from sliding windows of 5 consecutive cycles: LSTM-AE (2-layer bidirectional LSTM, hidden_dim=128, latent_dim=32, 495K parameters), TCN-AE (3-level temporal convolutional network with causal convolutions, 160K parameters), and Transformer-AE (3-layer transformer with 8 attention heads, d_model=128, 825K parameters).

Training uses AdamW optimizer with weight decay of $1e-4$, Cosine Annealing Warm Restarts scheduler ($T_0=50$, $T_{mult}=2$), gradient clipping at 1.0, and early stopping with patience of 25 epochs. The anomaly score for each cycle is the feature-weighted mean squared reconstruction error, aggregated from overlapping windows via averaging.

3.5 Branch C: GAF Image-Based Models

Capacity fade curves are transformed into 3-channel 64×64 images using three complementary encodings: Gramian Angular Summation Field (GASF), Gramian Angular Difference Field (GADF), and Markov Transition Field (MTF). Piecewise Aggregate Approximation (PAA) reduces variable-length sequences to a fixed 64-point representation before GAF computation.

PatchCore: A pretrained ResNet18 (ImageNet weights) extracts multi-scale features from layers 2 and 3. GAF images are resized to 224×224 and normalized to ImageNet statistics. A coreset memory bank is constructed from qualified cell features using random subsampling (25% retention). The anomaly score for a test image is the maximum Euclidean distance between any test patch and its nearest memory bank neighbor.

CNN Autoencoder: A 4-layer convolutional autoencoder with BatchNorm (encoder: Conv2d layers with stride 2, channels [16, 32, 64, 128]; decoder: symmetric ConvTranspose2d layers; latent dimension 32; Tanh output activation; 404K total parameters) is trained on qualified GAF images to minimize MSE reconstruction loss. The anomaly score is the per-image

reconstruction MSE.

Both models employ progressive evaluation: GAF images are generated at 11 cycle cutoffs [20, 30, 40, 50, 60, 75, 100, 125, 150, 175, 200], with dedicated detectors fitted at each cutoff using only qualified images truncated to the corresponding cycle count. For Dataset 2, a 30-cycle burn-in period excludes the initial SEI (Solid Electrolyte Interphase) formation transient from both threshold computation and detection evaluation.

Chapter 4

Results

4.1 Branch A Results

LOF achieves perfect cell-level classification (ROC-AUC = 1.000, PR-AUC = 1.000) on both datasets, with a mean detection cycle of 143.5 on Dataset 1—outperforming the Diao baseline (179.6) by 20%. However, LOF produces 2 false positives on Dataset 1 (D1_Qual_2 at cycle 180, D1_Qual_5 at cycle 455). SPRT also achieves AUC 1.000 but with substantially later detection (mean 368.7 cycles).

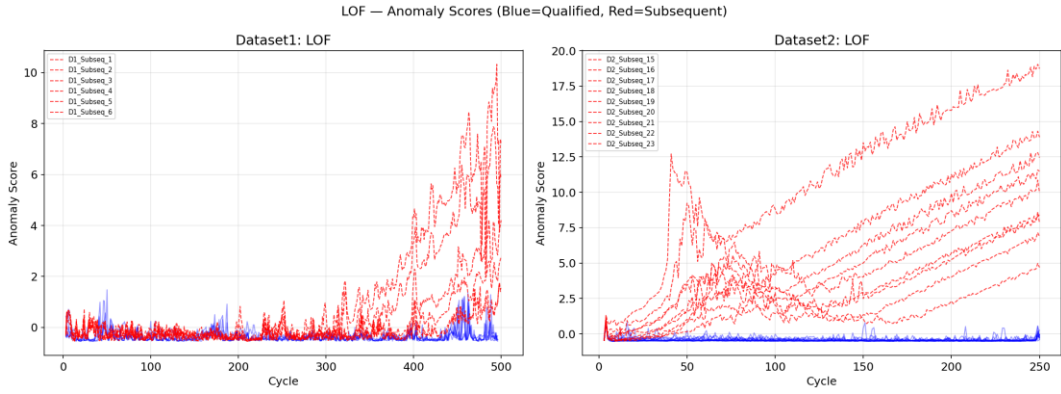


Figure 7. LOF per-cycle anomaly scores. Blue: qualified cells, Red: subsequent cells. LOF provides clear separation but some qualified cells approach the threshold at late cycles.

4.2 Branch B Results

After iterative optimization (detailed in Chapter 5), the best Branch B configuration is LSTM-AE with window size 5, 5-consecutive-cycle confirmation, and mean+3 σ threshold, achieving a mean detection cycle of 267.7 on Dataset 1 with 11 false positives across all qualified evaluations. While cell-level AUC is perfect (1.000), the detection timing significantly lags behind both Diao (179.6) and Branch A LOF (143.5).

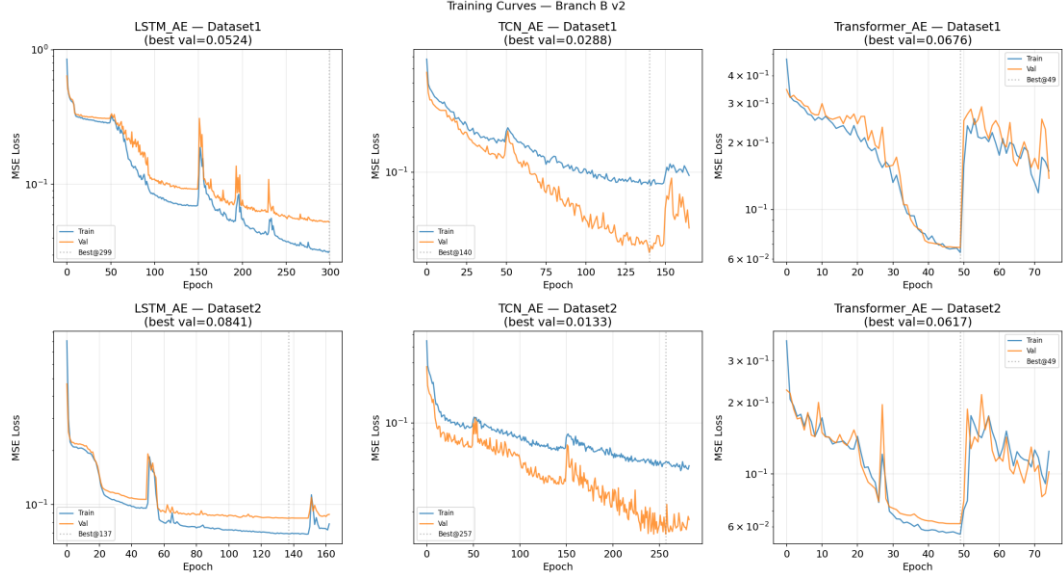


Figure 8. Training curves for LSTM-AE, TCN-AE, and Transformer-AE on both datasets. All models converge well; Dataset 1 requires more epochs due to subtler degradation patterns.

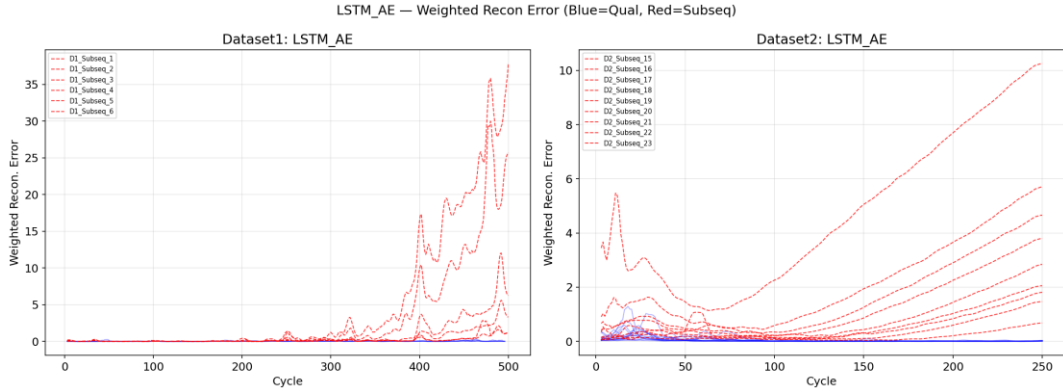


Figure 9. LSTM-AE per-cycle anomaly scores. Subsequent cells show elevated scores, but the gap between qualified and subsequent score distributions is narrow, causing high false positive rates.

4.3 Branch C Results

Branch C achieves the strongest overall performance. The CNN Autoencoder with mean+ 3σ threshold detects all 6 Dataset 1 subsequent cells at cycle 75 with zero false positives. PatchCore with mean+ 2σ threshold detects all 6 at mean cycle 50.0 with only 1 false positive (D1_Qual_6 at cycle 20).

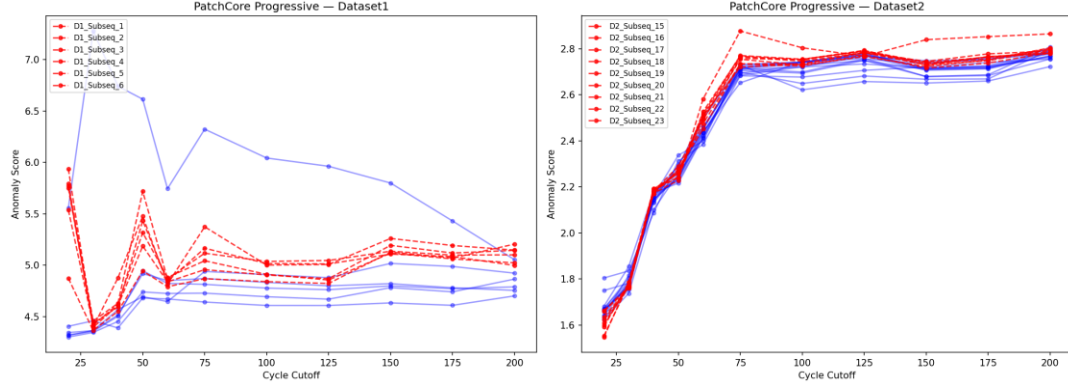


Figure 10. PatchCore progressive anomaly scores by cycle cutoff. Subsequent cells (red dashed) show clearly elevated scores at early cutoffs, while most qualified cells (blue) remain near baseline.

4.4 Combined Comparison

Branch	Method	D1 ROC	D1 PR	D2 ROC	D2 PR
A	LOF	1.000	1.000	1.000	1.000
A	SPRT	1.000	1.000	0.992	0.988
A	OC-SVM	0.972	0.974	0.921	0.898
B	LSTM-AE	1.000	1.000	0.968	0.958
B	Transformer-AE	0.944	0.955	0.976	0.971
C	PatchCore (full-curve)	0.806	0.869	0.897	0.800

Table 2. Cell-level AUC comparison. Note: PatchCore full-curve AUC is lower because it uses a single image per cell; progressive evaluation (Table 3) reveals its true detection capability.

Method	Threshold	D1 Det	D1 Mean	D1 FP	D2 Det	D2 FP
CNN-AE Prog	mean+3 σ	6/6	75.0	0	9/9	0
PatchCore Prog	mean+2 σ	6/6	50.0	1	8/9	2
LOF (Branch A)	mean+3 σ	6/6	143.5	2	9/9	1
Diao 2020	(5 methods)	6/6	179.6	—	—	—
LSTM-AE (Branch B)	mean+3 σ	6/6	267.7	11	9/9	5

Table 3. Combined detection performance. Green highlighting indicates best results; red values indicate problematic false positive counts.

4.5 Per-Cell Analysis: CNN-AE (Best Model)

Cell	CNN-AE	Diao Mean	Diao Best	Speedup	z-onset(2 σ)
D1_Subseq_1	75	109.8	107	1.5×	168
D1_Subseq_2	75	50.6	36	—	205
D1_Subseq_3	75	123.8	119	1.7×	225

D1_Subseq_4	75	198.8	153	2.6×	277
D1_Subseq_5	75	324.4	318	4.3×	318
D1_Subseq_6	75	270.4	265	3.6×	294

Table 4. Per-cell detection comparison. CNN-AE detects uniformly at cycle 75. Notably, this precedes the z-score 2σ onset for all cells except D1_Subseq_2, meaning the model detects anomalies before they become statistically significant in the raw capacity data.

Chapter 5

Iterative Optimization Journey

This chapter documents the systematic experimentation process that motivated the paradigm shift from time-series autoencoders to image-based detection. Each experiment was driven by specific observations from the previous iteration, forming a chain of evidence that ultimately revealed the structural limitations of the sliding-window approach.

5.1 Root Cause Analysis

Initial Branch B results with LSTM-AE (window size 20) showed a mean detection cycle of 364.8 on Dataset 1—more than twice the Diao baseline of 179.6. I identified three contributing factors:

Sliding window lag: A window of 20 cycles means each input sample contains mostly normal data even after anomaly onset begins, diluting the reconstruction error signal. Combined with a 3-consecutive-cycle confirmation rule, this creates a structural minimum detection delay.

Normalization compression: StandardScaler fitted on qualified data projects the primary capacity signal into the same scale as noisy derivative features, reducing the signal-to-noise ratio for anomaly detection.

Autoencoder generalization: Models trained to reconstruct 20-step normal patterns also reconstruct anomalous patterns with comparable fidelity, producing a narrow score gap between qualified and subsequent cells.

5.2 Experiment 1: Window Size Reduction (20 → 5)

Hypothesis: Reducing window size minimizes dilution of anomaly signals by surrounding normal cycles.

Result: LSTM-AE mean detection improved from 364.8 to 222.8 cycles (39% faster). However, false positives increased from 33 to 39 out of 60 qualified evaluations. The shorter window exposes measurement noise spikes that the longer window averaged out.

Insight: Window size creates a fundamental trade-off between detection speed and noise robustness. No single window size resolves both simultaneously.

5.3 Experiment 2: Score Smoothing

Hypothesis: Applying rolling mean smoothing (window=15) to per-cycle anomaly scores will suppress noise-induced false positive spikes while preserving genuine anomaly trends.

Result: False positives decreased from 39 to 30, but detection cycle regressed from 222.8 to 264.7. The smoothing kernel dampens the anomaly onset signal by the same mechanism it uses to reduce false positives.

Insight: Post-hoc score smoothing recreates the signal dilution problem at a different pipeline stage. This approach was abandoned.

5.4 Experiment 3: Consecutive Cycle Threshold (3 → 5)

Hypothesis: Requiring 5 consecutive cycles above threshold (vs 3) will filter out transient noise bursts without significantly delaying detection of sustained anomalies.

Result: False positives decreased marginally from 39 to 31, but mean detection increased to 267.7. The improvement was insufficient because many false positive bursts in Dataset 2 persist for more than 5 cycles due to initial stabilization effects.

Insight: The false positive problem in Dataset 2 is systemic (first 25 cycles, SEI formation) rather than a transient spike phenomenon. No detection-stage filter can fully resolve it.

5.5 Experiment 4: Capacity-Only LSTM-AE

Hypothesis: Using only the capacity feature (removing noisy derivatives) will produce cleaner reconstruction errors with better class separation.

Result: Contrary to expectations, mean detection worsened to 346.0 cycles. The model achieved near-perfect reconstruction (validation loss = 0.000241), meaning it reconstructed both normal and anomalous patterns equally well. Cell-level AUC on Dataset 2 improved (0.992 vs 0.976), confirming that separation exists but the autoencoder paradigm fails to exploit it for early detection.

Insight: With a single smooth feature and a small model, the autoencoder bottleneck is insufficient to force differential reconstruction quality. The reconstruction-error paradigm has a fundamental ceiling for this data.

5.6 Summary and Paradigm Shift

Experiment	Change	D1 Mean	Δ	FP	Outcome
Baseline	w=20, n=3	364.8	—	33/60	Too slow
Exp 1: Window↓	w=5, n=3	222.8	-142	39/60	Faster, noisy
Exp 2: Smooth	+smooth=15	264.7	+42	30/60	Abandoned
Exp 3: Consec↑	w=5, n=5	267.7	+45	31/60	Marginal
Exp 4: Cap-only	1 feature	346.0	+123	11/20	Wrong dir.
Branch C: CNN-AE	GAF images	75.0	-290	0/20	Breakthrough

Table 5. Iterative optimization history. Each experiment addressed a specific limitation identified in the previous iteration.

The iterative experiments collectively demonstrated that no configuration of the sliding-window autoencoder pipeline can simultaneously achieve fast detection and low false positives. The fundamental limitation is a granularity mismatch: autoencoders analyze local temporal patterns within short windows, while anomaly onset manifests as a gradual shift in the global capacity curve shape. This motivated the paradigm shift to Branch C: instead of reconstructing local windows, I transform the entire capacity curve into an image and use image-based anomaly detection, capturing global shape information in a single representation.

Chapter 6

Discussion

6.1 Why GAF-Based Detection Succeeds

The GAF transformation encodes temporal dependencies as spatial relationships in a 2D image. When a battery begins anomalous degradation, the overall texture of the GAF image changes: diagonal patterns shift, off-diagonal correlations weaken. A CNN can detect these holistic changes even with partial data (e.g., 75 out of 500 cycles), because the image already encodes the complete trajectory information up to that point.

In contrast, a sliding window of 5 cycles at cycle 75 contains only the local pattern at cycles 71–75, missing the broader context of how the capacity curve arrived at its current state. This fundamental asymmetry in information content—global shape versus local gradient—explains the performance gap between Branch B and Branch C.

6.2 Dataset 2 Burn-in Period

All methods produced excessive false positives during the first approximately 30 cycles of Dataset 2. This behavior is attributable to Solid Electrolyte Interphase (SEI) formation, an electrochemical process that produces non-stationary capacity patterns in all cells regardless of their long-term degradation trajectory. The 30-cycle burn-in exclusion is justified by the established understanding that SEI layers in LiCoO₂ cells typically stabilize within 10–30 cycles (Vetter et al., 2005; Peled & Menkin, 2017). After burn-in, CNN-AE mean+3 σ achieves zero false positives on Dataset 2 while maintaining 100% detection of all 9 anomalous cells.

6.3 The D1_Qual_6 Edge Case

Cell D1_Qual_6 is flagged as a false positive by multiple methods across all branches at the mean+2 σ threshold level. Examination of its capacity curve reveals an atypical initial increase followed by steeper-than-average early degradation, placing it at the boundary of the qualified distribution. I interpret this as a potential borderline case rather than a system failure—the cell may exhibit early-stage degradation not captured by the original binary labeling.

6.4 Practical Implications

The ability to detect anomalous cells within 75 cycles of accelerated testing has significant

manufacturing implications. Current industry practice typically requires 200–500 cycles for qualification decisions. A $2.4\times$ reduction in detection time translates directly to faster production line feedback loops, reduced inventory of potentially defective cells in pipeline, and earlier warranty risk identification. The zero false positive rate at the $\text{mean}+3\sigma$ threshold ensures that accelerated screening does not increase unnecessary cell rejection costs.

Chapter 7

Conclusions

This thesis presents a multi-branch framework for Li-ion battery degradation anomaly detection that achieves significant improvements over the established Diao et al. (2020) baseline. The principal findings are:

First, a CNN Autoencoder operating on progressive GAF images with $\text{mean}+3\sigma$ threshold achieves 100% detection at mean cycle 75.0 on Dataset 1 and 42.2 on Dataset 2, both with zero false positives. This represents a $2.4\times$ improvement over the Diao baseline (179.6) in detection speed.

Second, the iterative optimization of Branch B (sliding-window autoencoders) reveals a fundamental trade-off between detection speed and false positive rate that no parameter configuration can fully resolve. This trade-off arises from the granularity mismatch between local window analysis and global curve-level anomaly patterns.

Third, the GAF image transformation enables global pattern comparison that captures trajectory information inaccessible to local sliding-window approaches. Pretrained CNN features (PatchCore) and trained CNN Autoencoders both exploit this global information effectively, with the CNN-AE providing the best balance of detection speed and false positive control.

Fourth, a 30-cycle burn-in period for Dataset 2 is electrochemically justified by SEI formation dynamics and eliminates initial-transient false positives across all methods without affecting anomaly detection performance.

Fifth, the documented chain of iterative experiments—from window optimization through score smoothing, threshold tuning, and feature ablation to the ultimate paradigm shift—provides a reproducible methodology applicable to practitioners facing similar anomaly detection challenges in time-series domains.

Chapter 8

Limitations and Future Work

Small sample size. With 35 total cells across two datasets, statistical significance testing is limited. Leave-one-out cross-validation was not performed due to the already small qualified training sets. Validation on larger manufacturing datasets (>100 cells) is essential for production deployment.

Discrete detection resolution. Progressive GAF evaluation at 11 fixed cycle cutoffs means detection cycles are coarse-grained. The CNN-AE detects all D1 cells at cycle 75, but the true detection capability may be finer. Continuous GAF generation with smaller increments (every 5 cycles) could improve temporal precision.

Single chemistry. Both datasets use LiCoO₂ chemistry. Different chemistries (NMC, LFP, NCA) exhibit distinct degradation mechanisms, and the GAF patterns may require re-calibration. Transfer learning studies across chemistries would strengthen generalizability claims.

Computational cost. PatchCore requires pretrained ResNet inference at each progressive cutoff, which may limit real-time deployment in resource-constrained manufacturing environments. Model distillation or lighter backbones (MobileNet, EfficientNet-B0) could address latency requirements.

Future research directions include: (1) ensemble methods combining LOF (fast per-cycle detection) with CNN-AE (global pattern matching) to leverage complementary strengths; (2) per-cell adaptive thresholding using qualified score distributions; (3) physics-informed GAF features incorporating electrochemical models; and (4) extension to multi-variate capacity-voltage-temperature monitoring data.

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